Practical - 2

Practical Title: Semantic HTML5 Pages with Accessibility-Ready Structure

Problem Definition: Develop static HTML5 skeletons for at least 10 StudentHub pages such as

Home, About, Register, Login, Dashboard, Events, Profile, Contact,

Admin, FAQ, and Feedback. Use semantic tags and accessibility-friendly structure.

A. Theory :

1. Are semantic tags such as header, nav, main, section, article, aside, and footer used properly?
Answer:

No, they are only partially used.

Used correctly :- <nav>, <main>, and <section> are used properly across the pages.

Not used at all :- <header>, <footer>, <article>, and <aside> are missing from the entire folder. (e.g., The <nav> is not wrapped in a <header>, and there are no <footer> tags at the bottom of any page).

2. Are form labels, headings, image alt text, and navigation links accessible ?
Answer :

Yes . Form labels are correctly bound to inputs using for and id attributes (e.g., <label for="enrollment">). Headings (<h1>, <h2>) and navigation links (<a>) are used semantically.

3. Is the page structure consistent across all pages?
Answer :

Yes . The page structure is perfectly consistent.

Every page utilizes the exact same <nav class="navbar"> block at the top, followed by a <main> container for the core content.

B. Implementation :

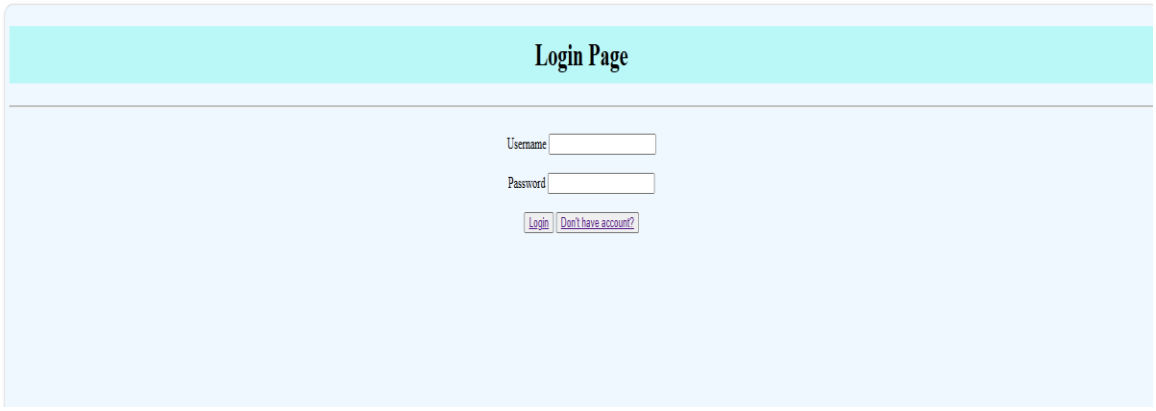
In this practical, the foundational structure of the StudentHub portal was developed.

Ten separate HTML pages (such as dashboard, courses, login, and result) were coded using core HTML5 elements.

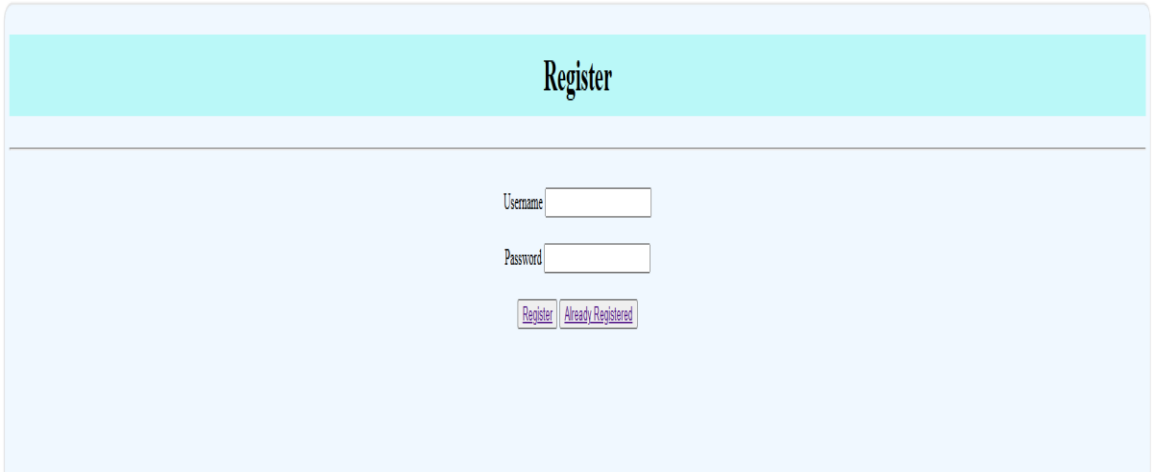
Semantic tags, complex <table> structures for data presentation, and <form> elements for user input were implemented to build the raw, unstyled framework of the application.

C. Evidence of Work (Screenshots) :

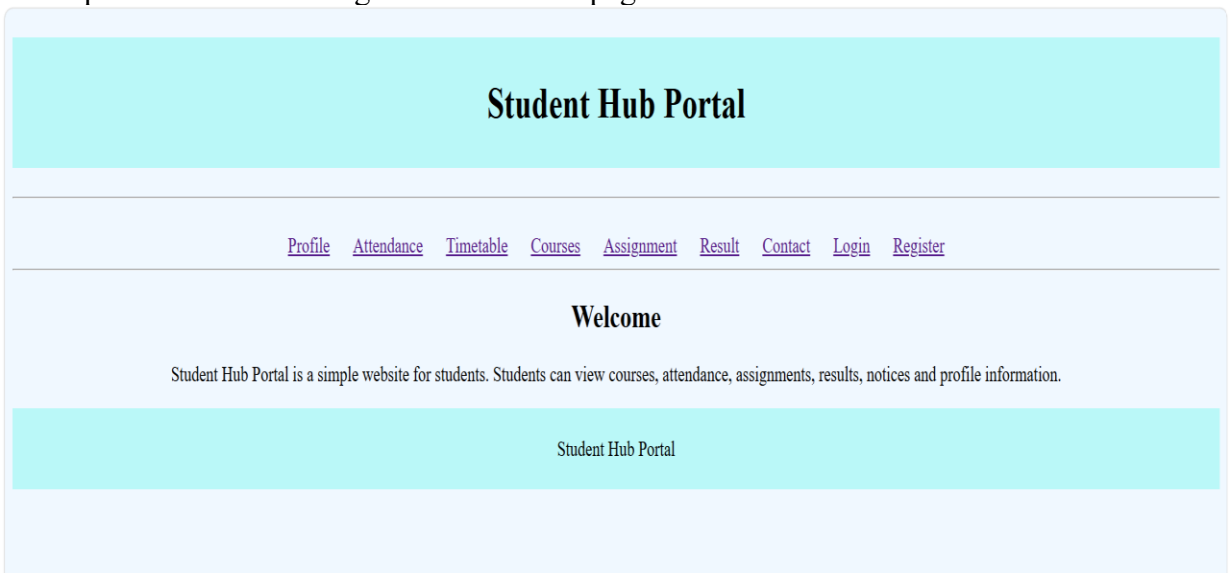
Description: wireframe image of login page



Description: wireframe image of Register page



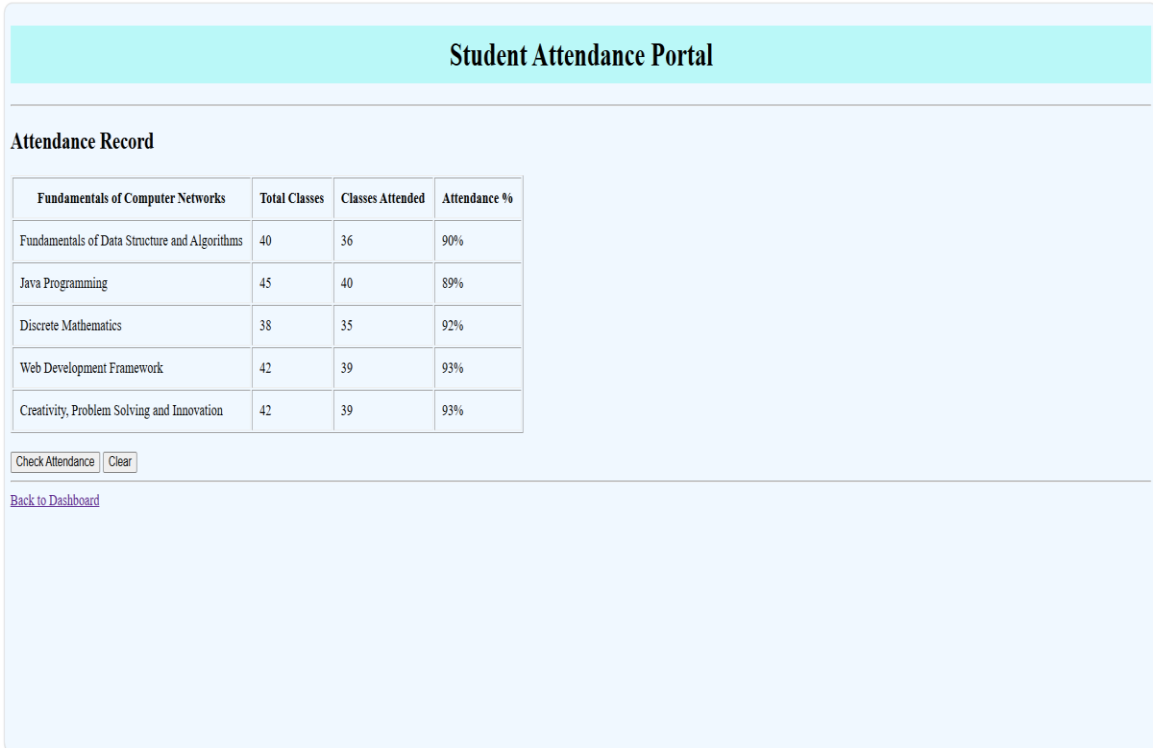
Description: wireframe image of Dashboard page



Description: wireframe image of Profile page



Description: wireframe image of Attendance page



Student Attendance Portal

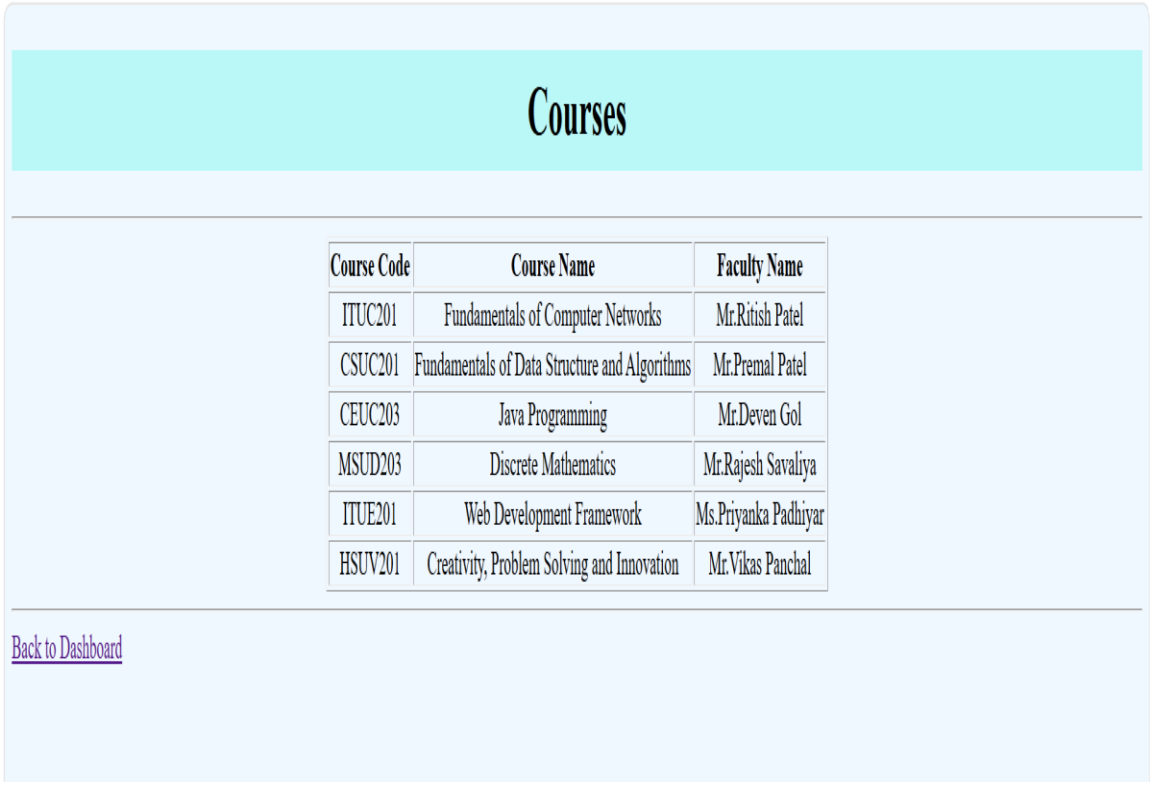
Attendance Record

Fundamentals of Computer Networks	Total Classes	Classes Attended	Attendance %
Fundamentals of Data Structure and Algorithms	40	36	90%
Java Programming	45	40	89%
Discrete Mathematics	38	35	92%
Web Development Framework	42	39	93%
Creativity, Problem Solving and Innovation	42	39	93%

Check Attendance Clear

Back to Dashboard

Description: wireframe image of Course page

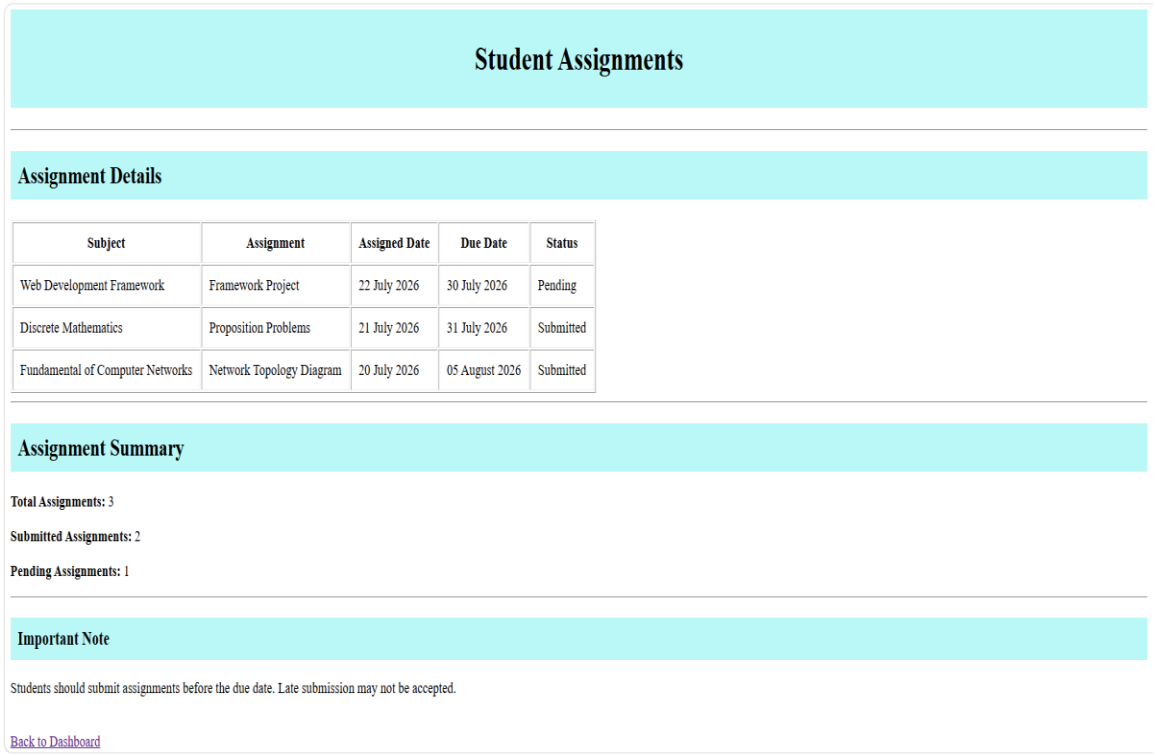


Courses

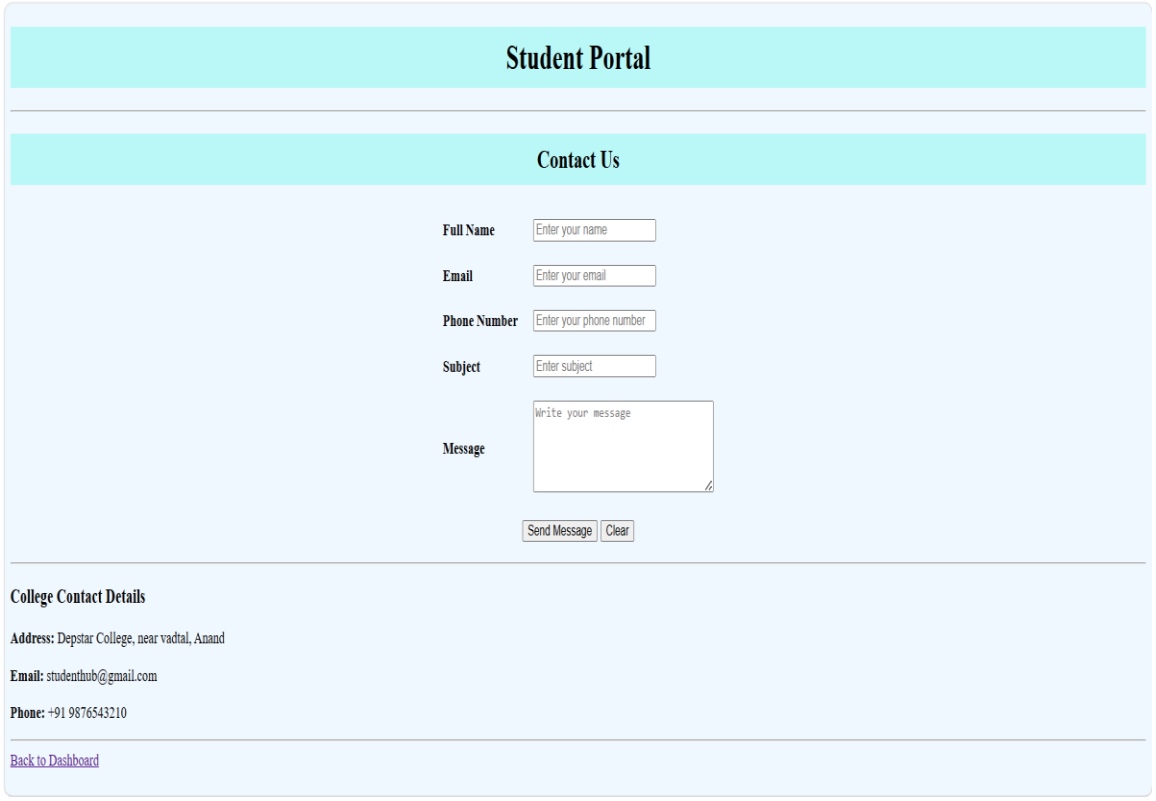
Course Code	Course Name	Faculty Name
ITUC201	Fundamentals of Computer Networks	Mr.Ritish Patel
CSUC201	Fundamentals of Data Structure and Algorithms	Mr.Premal Patel
CEUC203	Java Programming	Mr.Deven Gol
MSUD203	Discrete Mathematics	Mr.Rajesh Savaliya
ITUE201	Web Development Framework	Ms.Priyanka Padhiyar
HSUV201	Creativity, Problem Solving and Innovation	Mr.Vikas Panchal

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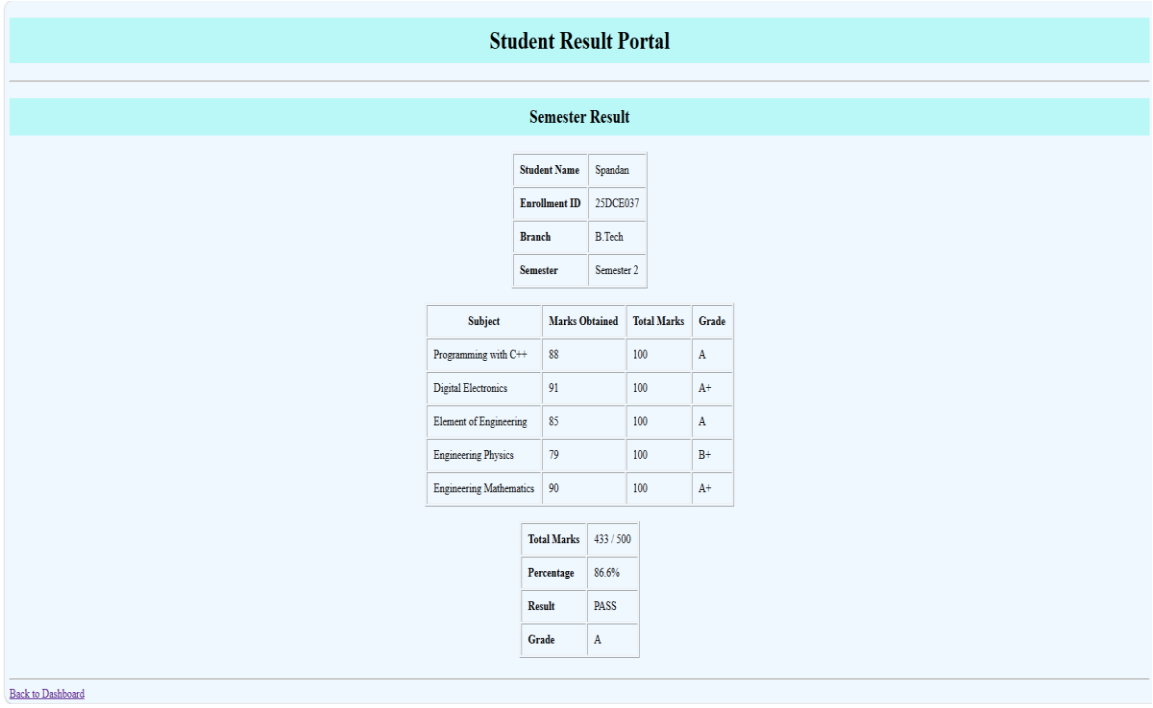
Description: wireframe image of Assignment page



Description: wireframe image of Contact page



Description: wireframe image of Result page



Student Result Portal

Semester Result

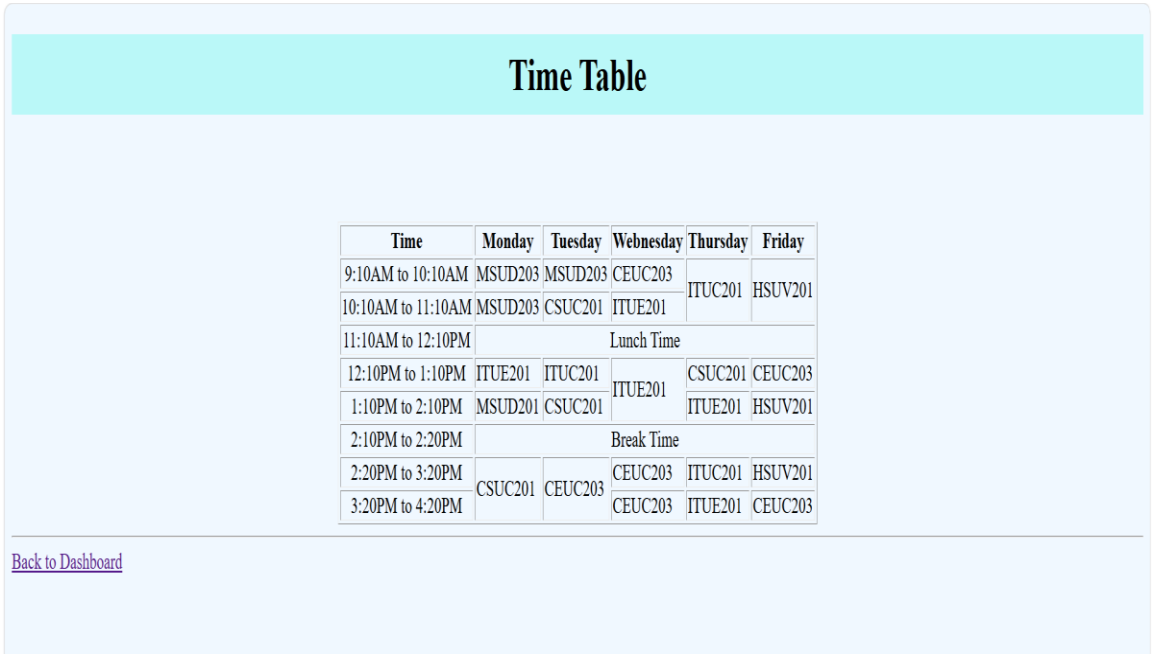
Student Name	Spandan
Enrollment ID	25DCE037
Branch	B.Tech
Semester	Semester 2

Subject	Marks Obtained	Total Marks	Grade
Programming with C++	88	100	A
Digital Electronics	91	100	A+
Element of Engineering	85	100	A
Engineering Physics	79	100	B+
Engineering Mathematics	90	100	A+

Total Marks	433 / 500
Percentage	86.6%
Result	PASS
Grade	A

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Description: wireframe image of Time Table page



Time Table

Time	Monday	Tuesday	Webnesday	Thursday	Friday
9:10AM to 10:10AM	MSUD203	MSUD203	CEUC203	ITUC201	HSUV201
10:10AM to 11:10AM	MSUD203	CSUC201	ITUE201		
11:10AM to 12:10PM	Lunch Time				
12:10PM to 1:10PM	ITUE201	ITUC201	ITUE201	CSUC201	CEUC203
1:10PM to 2:10PM	MSUD201	CSUC201		ITUE201	HSUV201
2:10PM to 2:20PM	Break Time				
2:20PM to 3:20PM	CSUC201	CEUC203	CEUC203	ITUC201	HSUV201
3:20PM to 4:20PM			CEUC203	ITUE201	CEUC203

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D. GitHub Repository Link:

https://github.com/25dce037-tech/Student-Hub-Portal.git

E. Challenges Faced:

During Practical 2, the main challenge was building the portal strictly using unstyled HTML elements while ensuring the content remained logical and readable.

Creating complex pages like the results and dashboard required careful nesting of <table>, <tr>, and <td> tags to present data clearly.

Another hurdle was designing the forms for login and registration, ensuring all <input> fields had the correct attributes for future backend integration.

Furthermore, manually linking 10 distinct files using <a href="filename.html"> tags meant that testing the navigation flow and preventing dead links was a tedious and error-prone process.

F. Learning Outcomes:

Mastered the creation of structured, multi-page HTML5 documents

Learned how to build complex data tables and interactive user input forms.

Gained experience using semantic HTML to ensure clear document structure.

Understood how to manually link static web pages to form a cohesive portal.

Faculty Remarks: